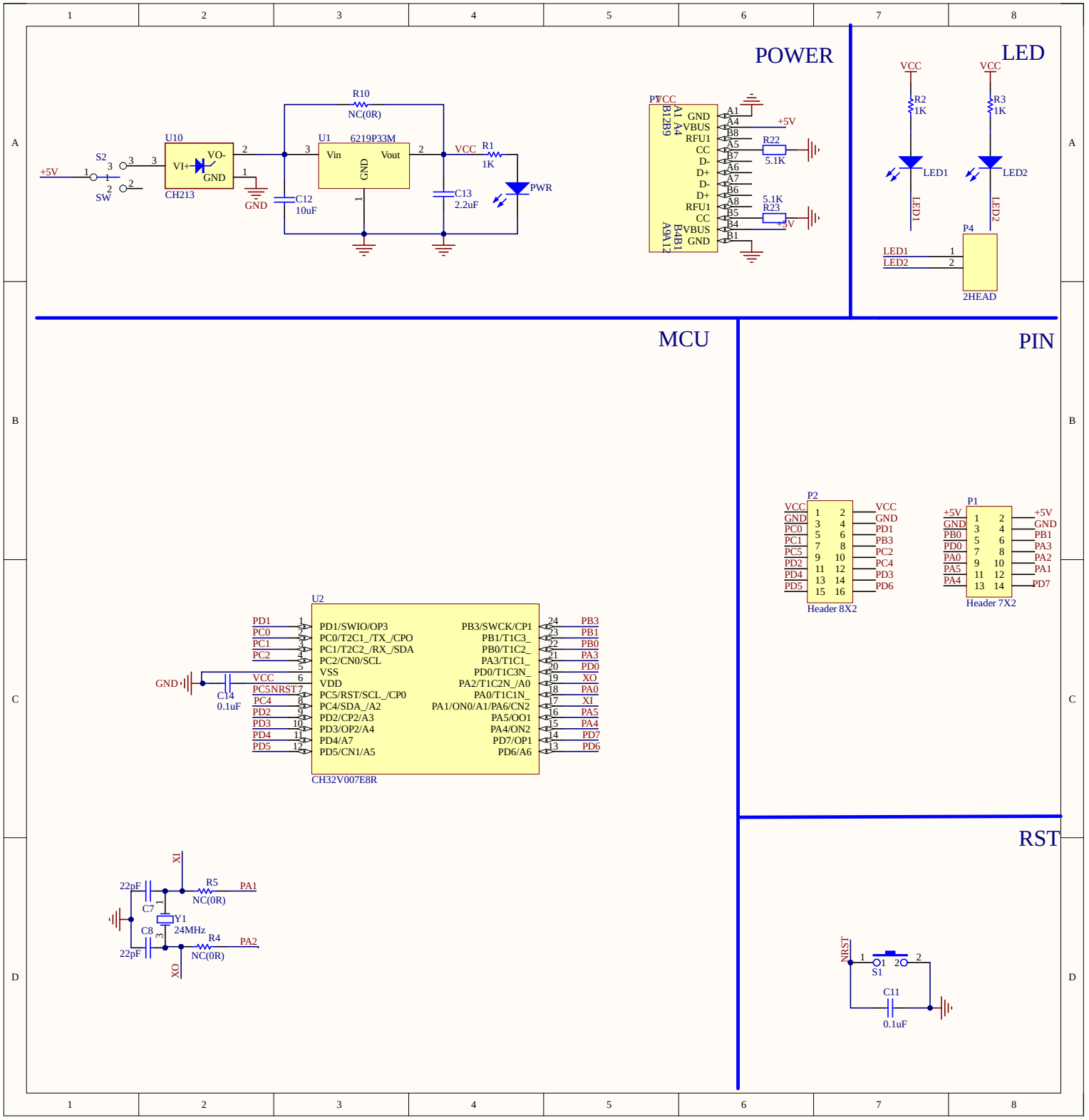
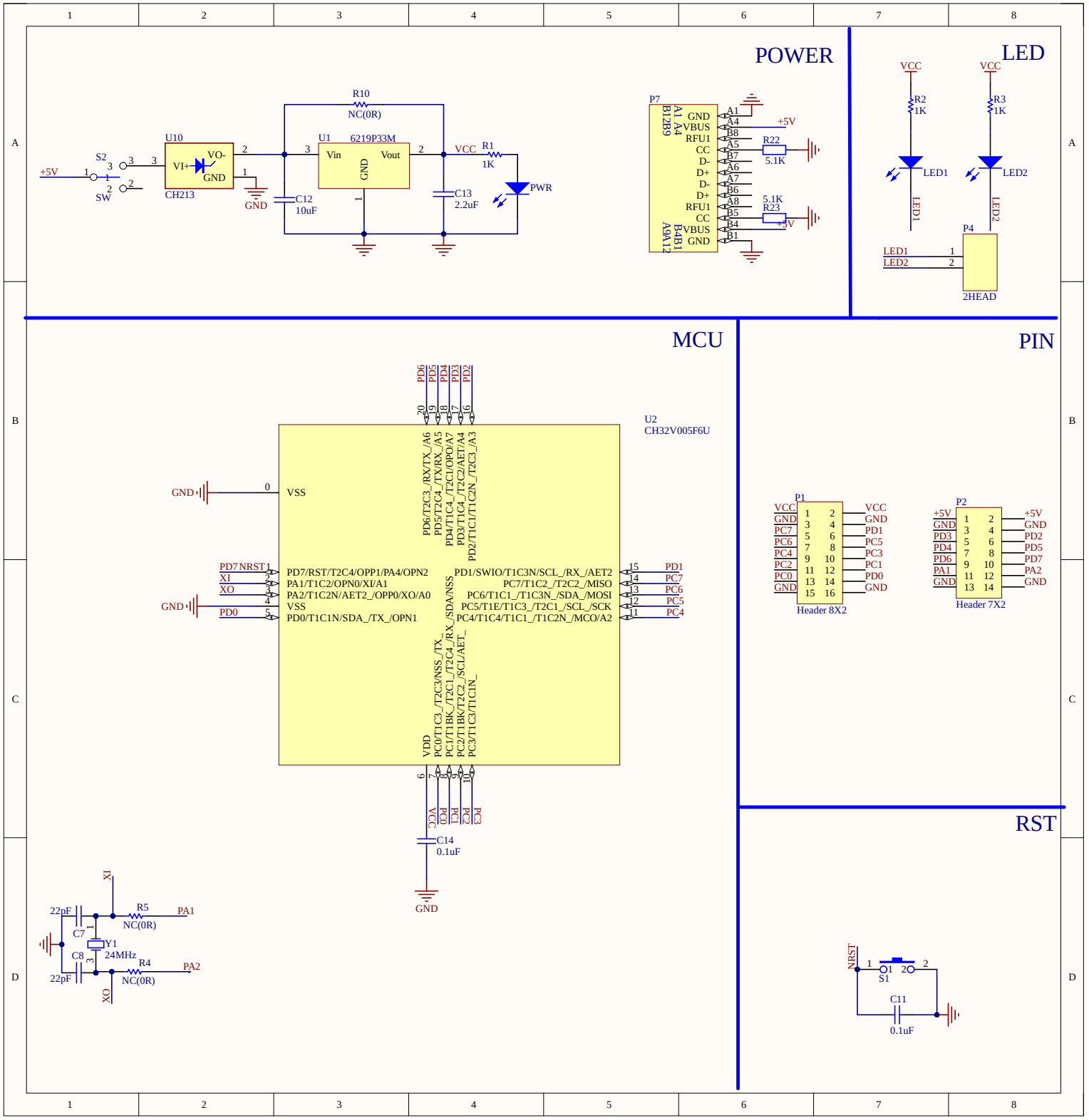
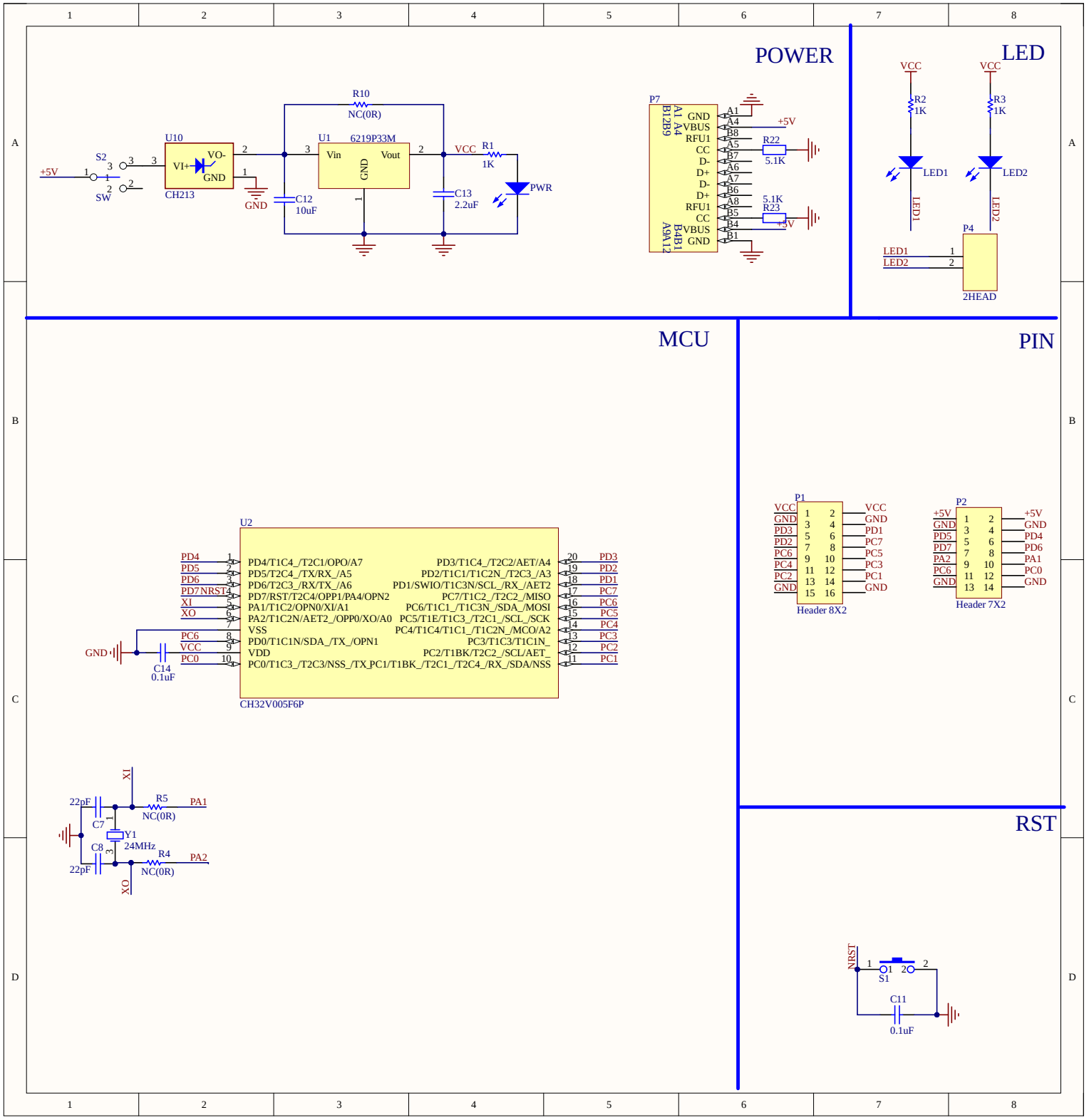
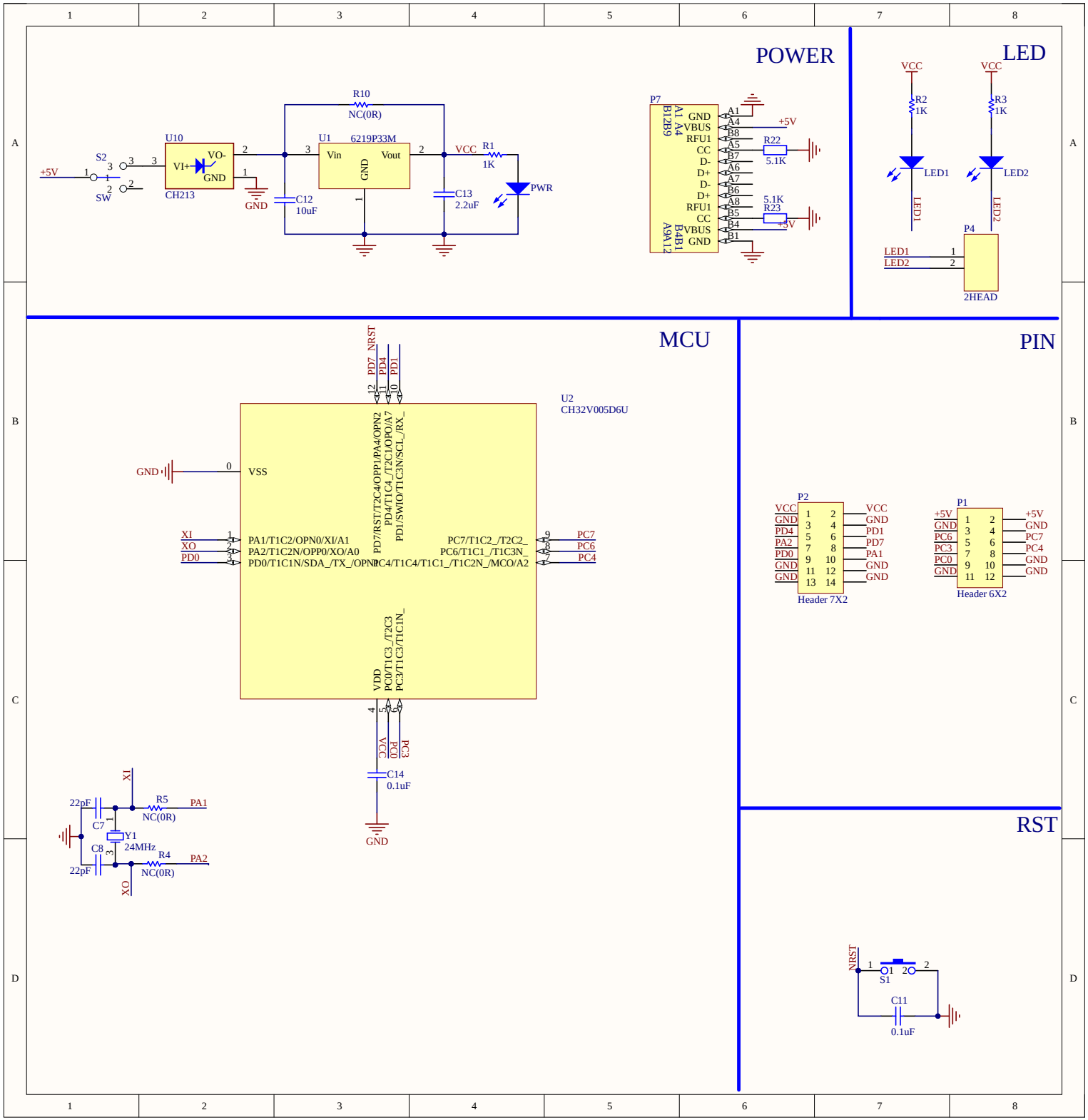
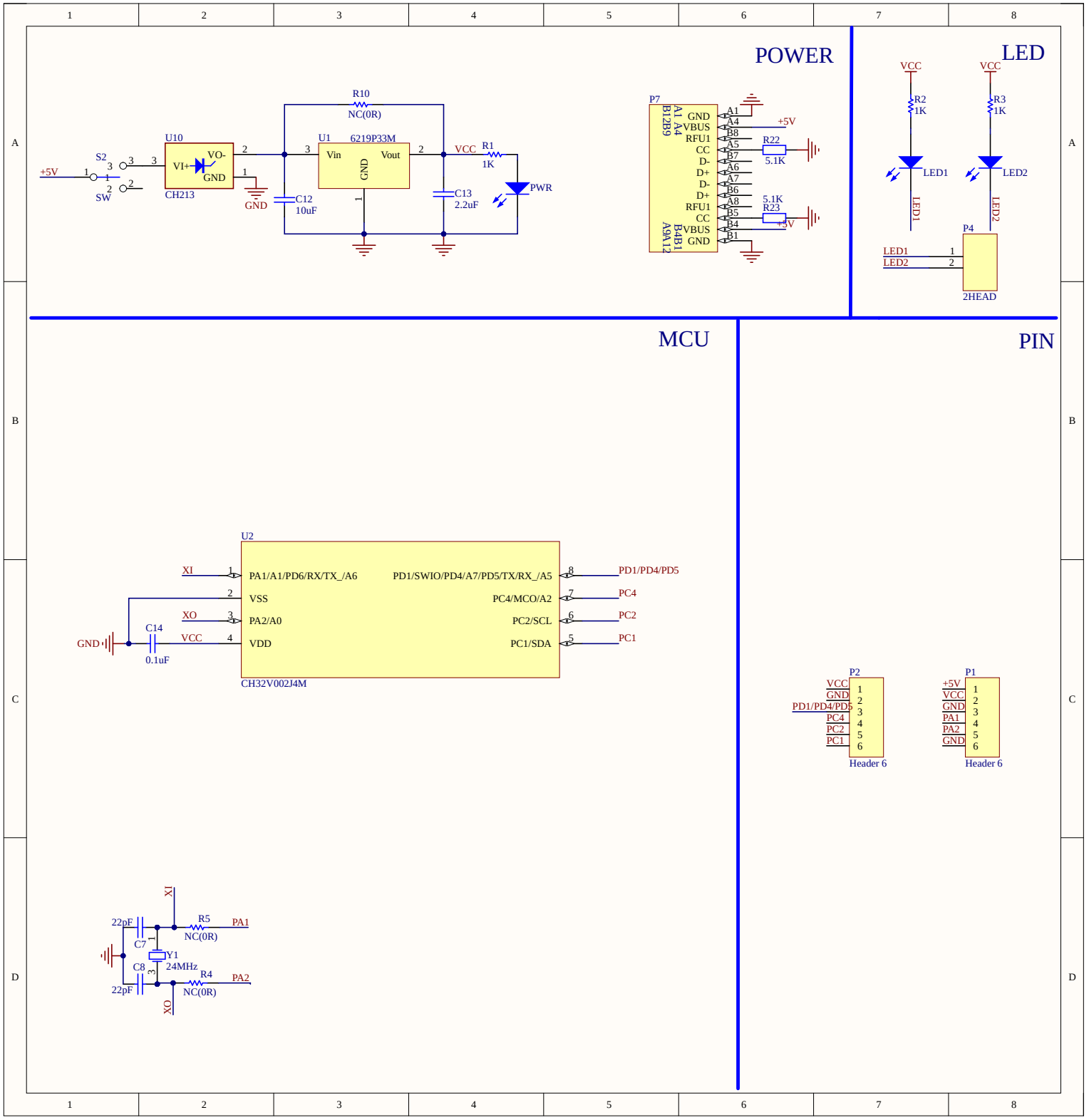
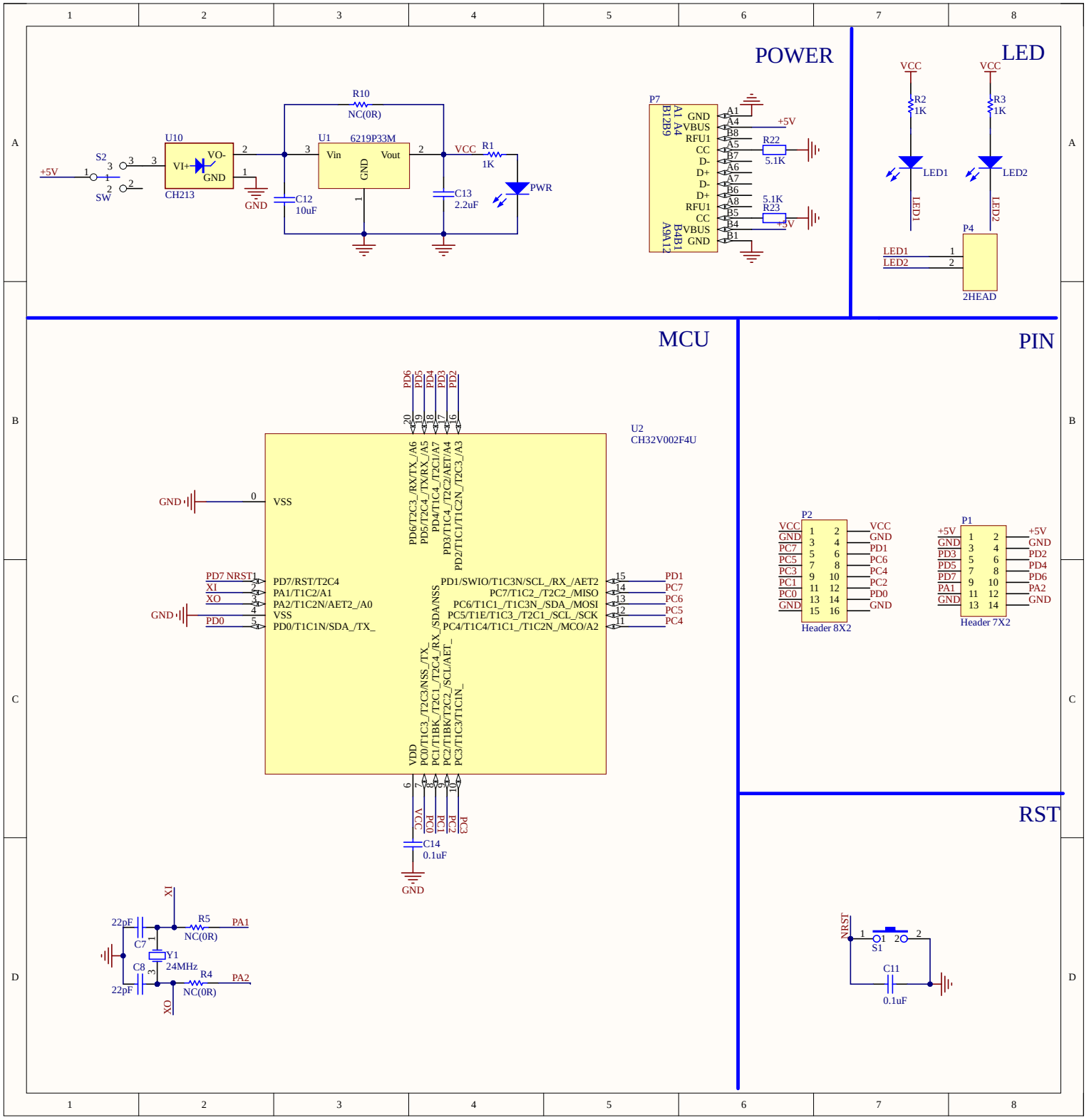

POWER
LED
MCU
PIN
RST
U10
CH213
U1
6219P33M
R10
NC(0R)
R1
1K
C12
10uF
C13
2.2uF
PWR
S2
SW
+5V
GND
VCC
R22
5.1K
R23
5.1K
R2
1K
R3
1K
LED1
LED2
P4
2HEAD
U2
CH32V007E8R
PD1/SWIO/OP3
PC0/T2C1_/TX_/CPO
PC1/T2C2_/RX_/SDA
PC2/CN0/SCL
VSS
VDD
PC5/RST/SCL_/CP0
PC4/SDA_/A2
PD2/CP2/A3
PD3/OP2/A4
PD4/A7
PD5/CN1/A5
PB3/SWCK/CP1
PB1/T1C3_
PB0/T1C2_
PA3/T1C1_
PD0/T1C3N_
PA2/T1C2N_/A0
PA0/T1C1N_
PA1/ON0/A1/PA6/CN2
PA5/OO1
PA4/ON2
PD7/OP1
PD6/A6
C14
0.1uF
P2
Header 8X2
P1
Header 7X2
22pF
C7
C8
Y1
24MHz
R5
R4
NC(0R)
XI
XO
PA1
PA2
NRST
S1
C11
0.1uF

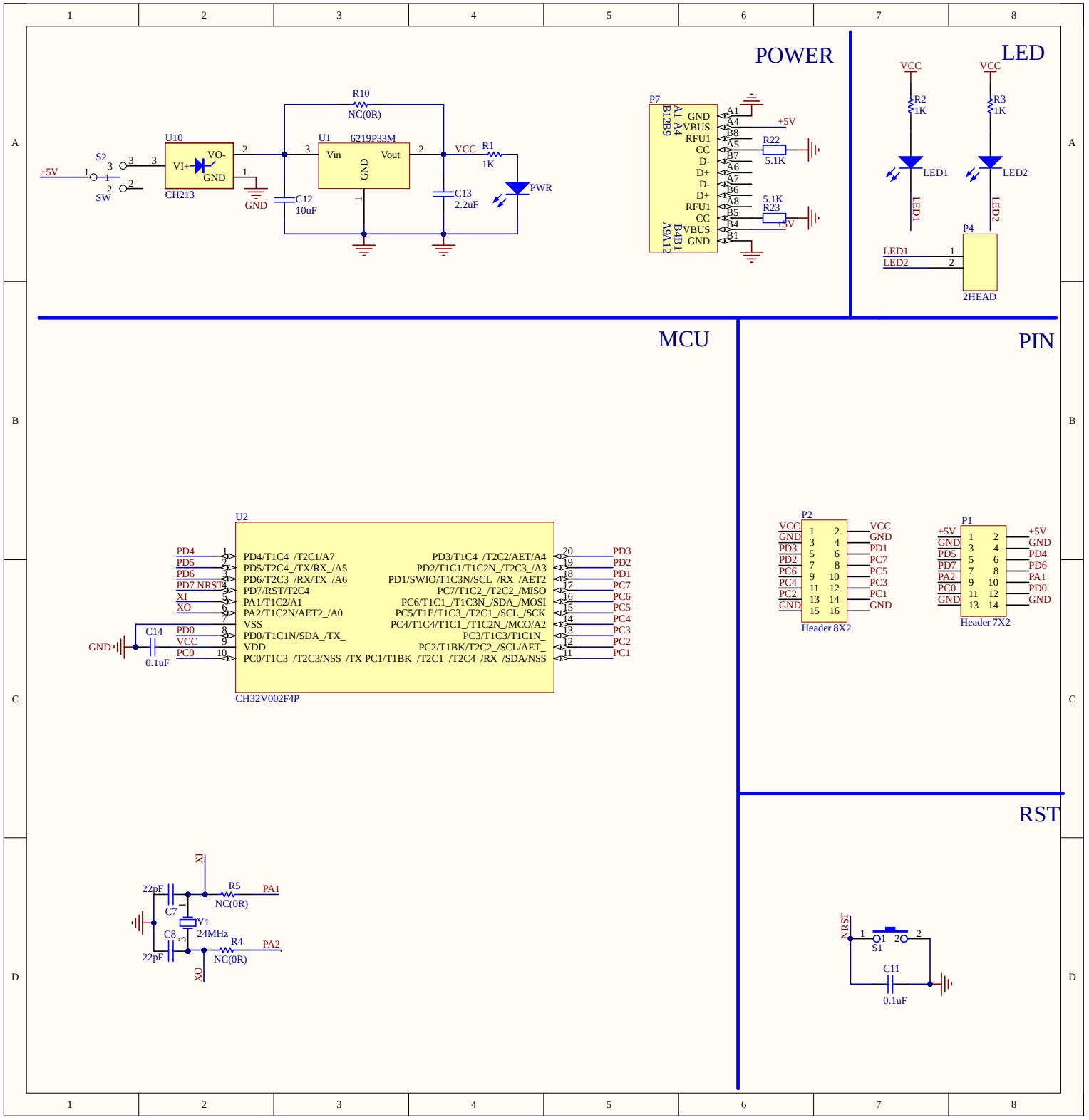

POWER
LED
MCU
PIN
RST
U10
CH213
U1
6219P33M
U2
CH32V005F6U
P1
P2
P4
P7
Header 8X2
Header 7X2
2HEAD
LED1
LED2
PWR
R10
NC(0R)
R1
1K
R2
1K
R3
1K
R22
5.1K
R23
5.1K
C12
10uF
C13
2.2uF
C14
0.1uF
C11
0.1uF
C7
C8
22pF
22pF
Y1
24MHz
R4
NC(0R)
R5
NC(0R)
S1
S2
SW
+5V
GND
VCC
PA1
PA2
XI
XO
NRST

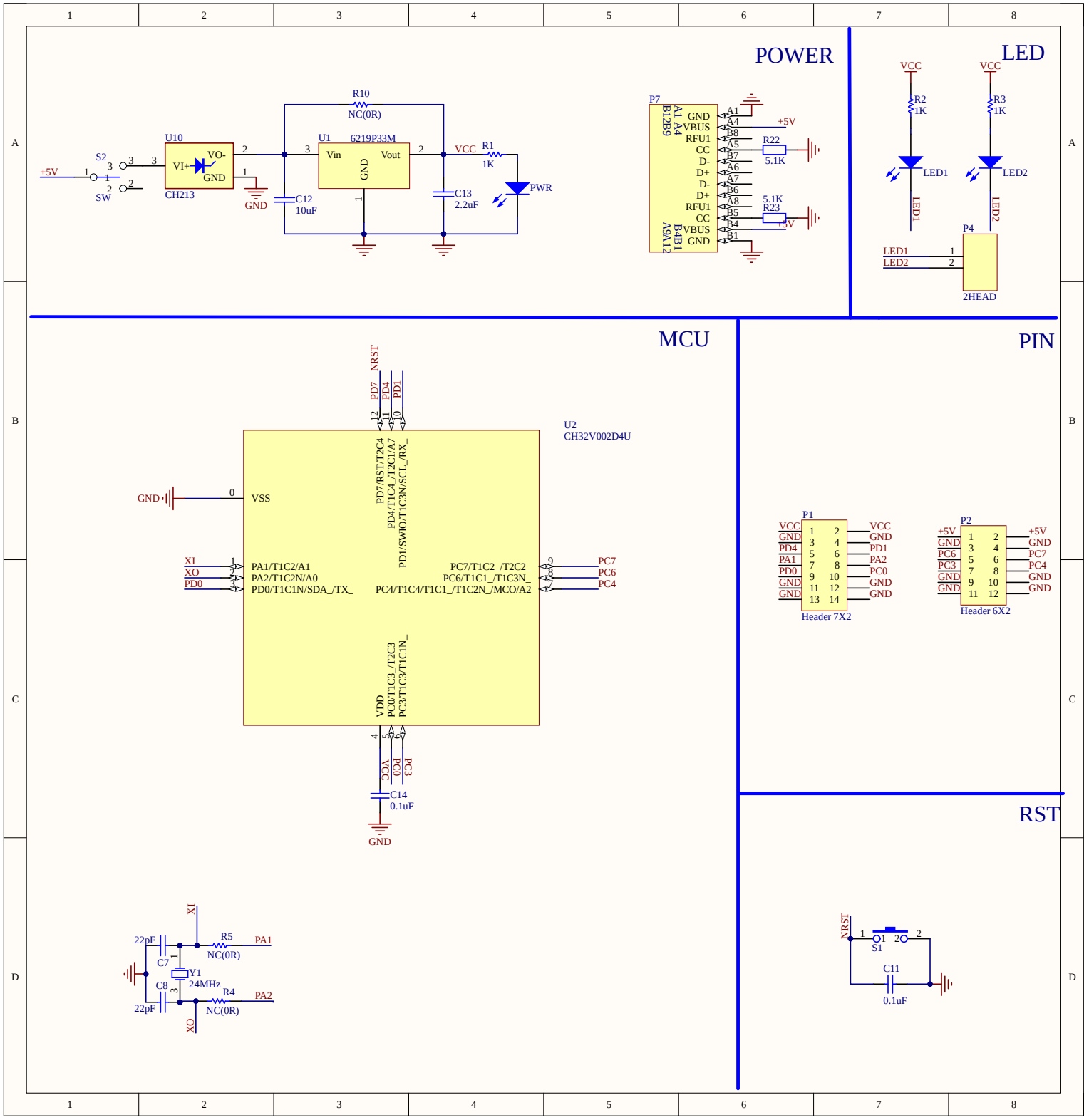

POWER
LED
MCU
PIN
RST
R10
NC(0R)
U10
VO-
VI+
GND
CH213
U1
6219P33M
Vin
Vout
GND
VCC
R1
1K
PWR
C12
10uF
C13
2.2uF
S2
SW
+5V
P7
GND
VBUS
RFU1
CC
D-
D+
D-
D+
RFU1
CC
VBUS
GND
R22
5.1K
R23
5.1K
+5V
R2
1K
R3
1K
LED1
LED2
P4
2HEAD
U2
PD4
PD5
PD6
PD7NRST
XI
XO
PC6
VCC
PC0
PD4/T1C4_/T2C1/OPO/A7
PD5/T2C4_/TX/RX_/A5
PD6/T2C3_/RX/TX_/A6
PD7/RST/T2C4/OPP1/PA4/OPN2
PA1/T1C2/OPN0/XI/A1
PA2/T1C2N/AET2_/OPP0/XO/A0
VSS
PD0/T1C1N/SDA_/TX_/OPN1
VDD
PC0/T1C3_/T2C3/NSS_/TX
PD3/T1C4_/T2C2/AET/A4
PD2/T1C1/T1C2N_/T2C3_/A3
PD1/SWIO/T1C3N/SCL_/RX_/AET2
PC7/T1C2_/T2C2_/MISO
PC6/T1C1_/T1C3N_/SDA_/MOSI
PC5/T1E/T1C3_/T2C1_/SCL_/SCK
PC4/T1C4/T1C1_/T1C2N_/MCO/A2
PC3/T1C3/T1C1N_
PC2/T1BK/T2C2_/SCL/AET_
PC1/T1BK_/T2C1_/T2C4_/RX_/SDA/NSS
CH32V005F6P
C14
0.1uF
P1
Header 8X2
P2
Header 7X2
22pF
C7
Y1
24MHz
C8
22pF
R5
NC(0R)
R4
NC(0R)
NRST
S1
C11
0.1uF

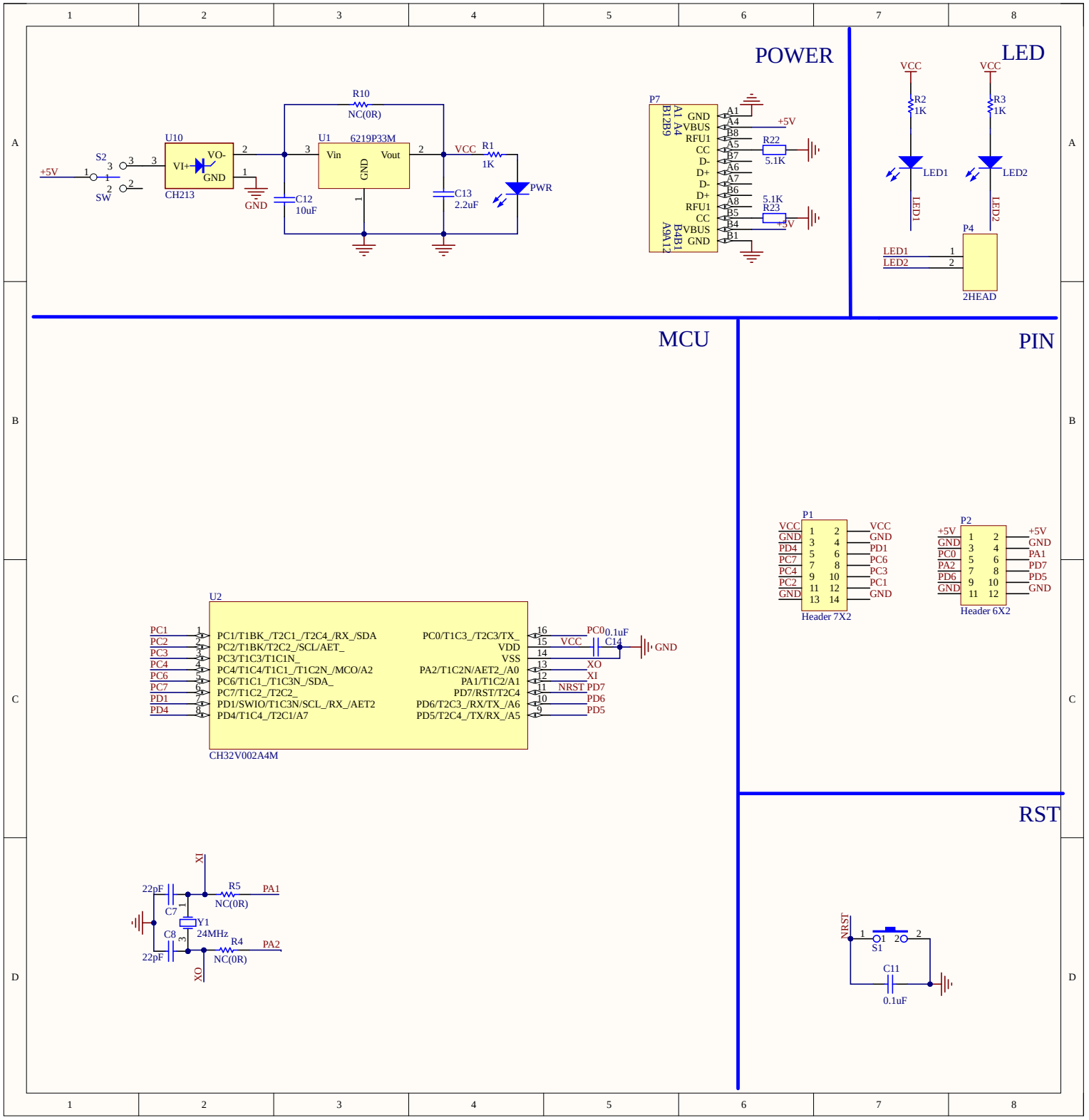

POWER
LED
MCU
PIN
RST
R10
NC(0R)
U10
CH213
U1
6219P33M
Vin
Vout
GND
VCC
R1
1K
PWR
C12
10uF
C13
2.2uF
S2
SW
+5V
P7
R22
5.1K
R23
5.1K
R2
1K
R3
1K
LED1
LED2
P4
2HEAD
U2
CH32V005D6U
VSS
VDD
NRST
XI
XO
PA1/T1C2/OPN0/XI/A1
PA2/T1C2N/OPP0/XO/A0
PD0/T1C1N/SDA_/TX_/OPN1
PD7/RST/T2C4/OPP1/PA4/OPN2
PD4/T1C4_/T2C1/OPO/A7
PD1/SWIO/T1C3N/SCL_/RX_
PC7/T1C2_/T2C2_
PC6/T1C1_/T1C3N_
PC4/T1C4/T1C1_/T1C2N_/MCO/A2
PC0/T1C3_/T2C3
PC3/T1C3/T1C1N_
C14
0.1uF
C7
C8
22pF
Y1
24MHz
R5
R4
NC(0R)
PA1
PA2
P2
Header 7X2
P1
Header 6X2
S1
C11
0.1uF

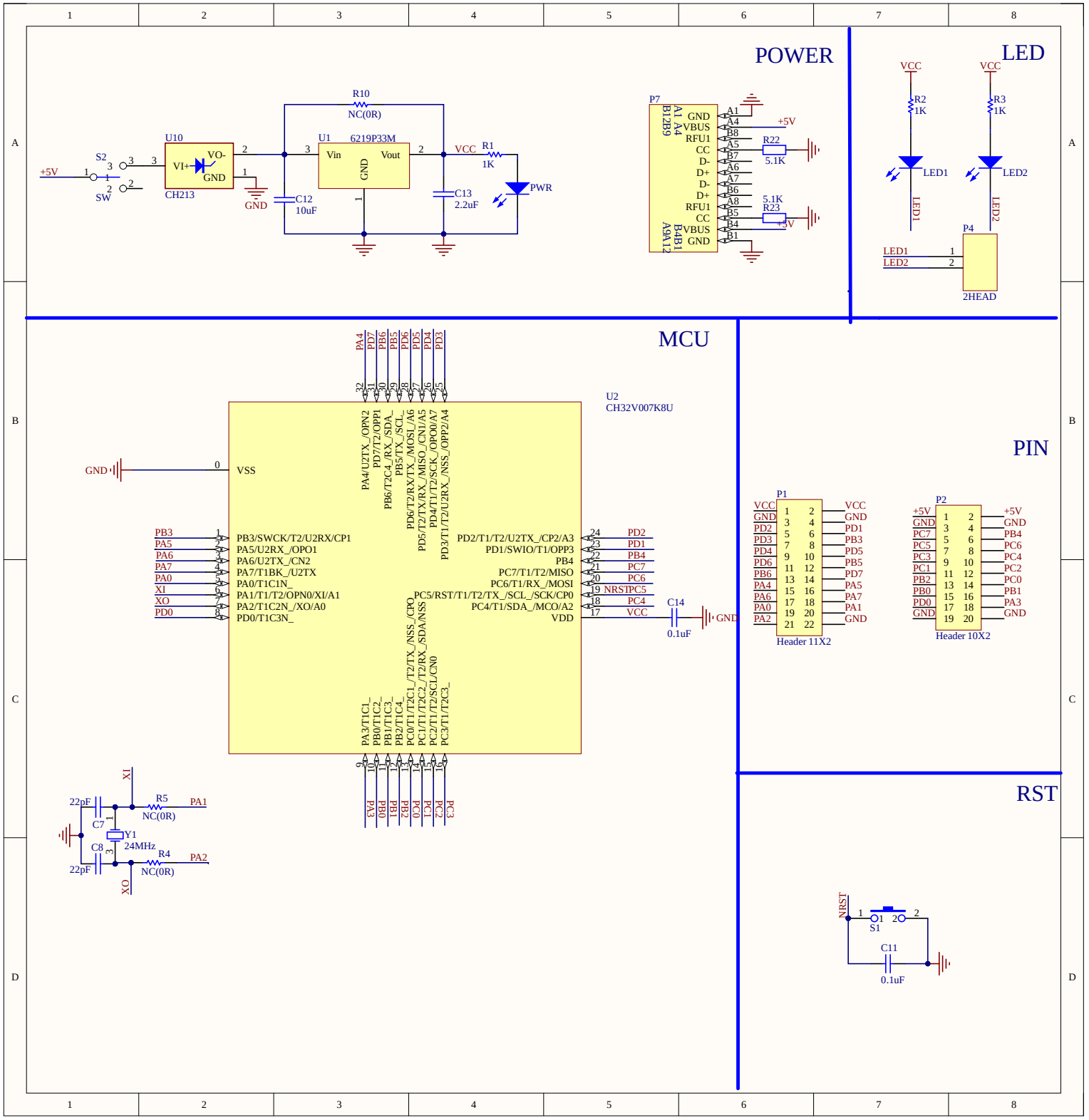

POWER
LED
MCU
PIN
R10
NC(0R)
U10
VO-
VI+
GND
CH213
S2
SW
+5V
C12
10uF
U1
6219P33M
Vin
Vout
GND
VCC
R1
1K
C13
2.2uF
PWR
P7
GND
VBUS
RFU1
CC
D-
D+
D-
D+
RFU1
CC
VBUS
GND
R22
5.1K
R23
5.1K
+5V
VCC
R2
1K
R3
1K
LED1
LED2
P4
2HEAD
U2
XI
XO
VSS
PA1/A1/PD6/RX/TX_/A6
PA2/A0
VDD
PD1/SWIO/PD4/A7/PD5/TX/RX_/A5
PC4/MCO/A2
PC2/SCL
PC1/SDA
PD1/PD4/PD5
PC4
PC2
PC1
C14
0.1uF
CH32V002J4M
P2
P1
Header 6
Header 6
C7
C8
22pF
22pF
Y1
24MHz
R5
NC(0R)
R4
NC(0R)
PA1
PA2

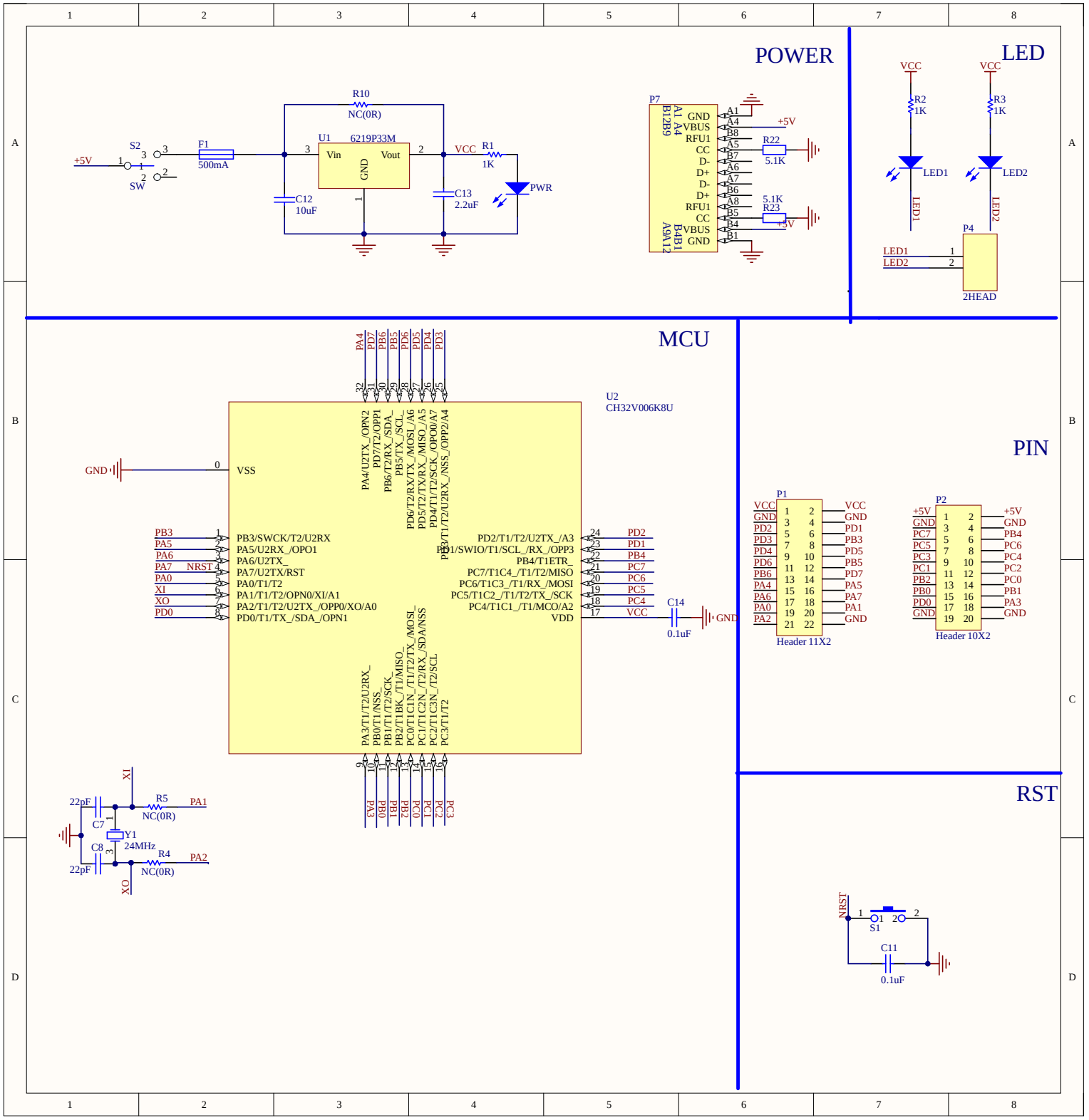

POWER
LED
MCU
PIN
RST
U10
CH213
U1
6219P33M
R10
NC(0R)
R1
1K
C12
10uF
C13
2.2uF
PWR
S2
SW
+5V
GND
VCC
P7
R22
5.1K
R23
5.1K
R2
1K
R3
1K
LED1
LED2
P4
2HEAD
U2
CH32V002F4U
VSS
VDD
PD7 NRST
XI
XO
PD0
PD1
PC7
PC6
PC5
PC4
C14
0.1uF
C7
C8
22pF
Y1
24MHz
R5
R4
NC(0R)
PA1
PA2
P2
Header 8X2
P1
Header 7X2
NRST
S1
C11
0.1uF


POWER
LED
MCU
PIN
RST
U10
CH213
U1
6219P33M
R10
NC(0R)
R1
1K
C12
10uF
C13
2.2uF
PWR
S2
SW
+5V
GND
VCC
P7
R22
5.1K
R23
5.1K
R2
1K
R3
1K
LED1
LED2
P4
2HEAD
U2
CH32V002F4P
PD4/T1C4_/T2C1/A7
PD5/T2C4_/TX/RX_/A5
PD6/T2C3_/RX/TX_/A6
PD7/RST/T2C4
PA1/T1C2/A1
PA2/T1C2N/AET2_/A0
VSS
PD0/T1C1N/SDA_/TX_
VDD
PC0/T1C3_/T2C3/NSS_/TX
PD3/T1C4_/T2C2/AET/A4
PD2/T1C1/T1C2N_/T2C3_/A3
PD1/SWIO/T1C3N/SCL_/RX_/AET2
PC7/T1C2_/T2C2_/MISO
PC6/T1C1_/T1C3N_/SDA_/MOSI
PC5/T1E/T1C3_/T2C1_/SCL_/SCK
PC4/T1C4/T1C1_/T1C2N_/MCO/A2
PC3/T1C3/T1C1N_
PC2/T1BK/T2C2_/SCL/AET_
PC1/T1BK_/T2C1_/T2C4_/RX_/SDA/NSS
C14
0.1uF
P2
Header 8X2
P1
Header 7X2
XI
XO
R5
NC(0R)
R4
NC(0R)
C7
C8
Y1
24MHz
22pF
PA1
PA2
NRST
S1
C11
0.1uF


POWER
LED
MCU
PIN
RST
R10
NC(0R)
U10
VO-
VI+
GND
CH213
S2
SW
+5V
U1
6219P33M
Vin
Vout
GND
VCC
R1
1K
C12
10uF
C13
2.2uF
PWR
P7
GND
VBUS
RFU1
CC
D-
D+
R22
5.1K
R23
5.1K
R2
1K
R3
1K
LED1
LED2
P4
2HEAD
U2
CH32V002D4U
VSS
PA1/T1C2/A1
PA2/T1C2N/A0
PD0/T1C1N/SDA_/TX_
PD7/RST/T2C4
PD4/T1C4_/T2C1/A7
PD1/SWIO/T1C3N/SCL_/RX_
PC7/T1C2_/T2C2_
PC6/T1C1_/T1C3N_
PC4/T1C4/T1C1_/T1C2N_/MCO/A2
VDD
PC0/T1C3_/T2C3
PC3/T1C3/T1C1N_
C14
0.1uF
XI
XO
NRST
PD7
PD4
PD1
PC7
PC6
PC4
PC0
PC3
VCC
R5
NC(0R)
R4
NC(0R)
C7
C8
Y1
24MHz
22pF
22pF
PA1
PA2
P1
Header 7X2
P2
Header 6X2
S1
C11
0.1uF


POWER
LED
MCU
PIN
RST
R10
NC(0R)
U10
VO-
VI+
GND
CH213
S2
SW
+5V
C12
10uF
U1
6219P33M
Vin
Vout
GND
C13
2.2uF
VCC
R1
1K
PWR
P7
GND
VBUS
RFU1
CC
D-
D+
RFU1
CC
VBUS
GND
R22
5.1K
R23
5.1K
R2
1K
R3
1K
LED1
LED2
P4
2HEAD
U2
PC1/T1BK_/T2C1_/T2C4_/RX_/SDA
PC2/T1BK/T2C2_/SCL/AET_
PC3/T1C3/T1C1N_
PC4/T1C4/T1C1_/T1C2N_/MCO/A2
PC6/T1C1_/T1C3N_/SDA_
PC7/T1C2_/T2C2_
PD1/SWIO/T1C3N/SCL_/RX_/AET2
PD4/T1C4_/T2C1/A7
PC0/T1C3_/T2C3/TX_
VDD
VSS
PA2/T1C2N/AET2_/A0
PA1/T1C2/A1
PD7/RST/T2C4
PD6/T2C3_/RX/TX_/A6
PD5/T2C4_/TX/RX_/A5
CH32V002A4M
C14
0.1uF
P1
Header 7X2
P2
Header 6X2
XI
XO
R5
NC(0R)
R4
NC(0R)
C7
C8
Y1
24MHz
22pF
22pF
PA1
PA2
NRST
S1
C11
0.1uF


POWER
LED
MCU
PIN
RST
R10
NC(0R)
U10
CH213
U1
6219P33M
Vin
Vout
GND
VCC
R1
1K
PWR
C12
10uF
C13
2.2uF
S2
SW
+5V
P7
R22
5.1K
R23
5.1K
R2
1K
R3
1K
LED1
LED2
P4
2HEAD
U2
CH32V007K8U
VSS
PB3/SWCK/T2/U2RX/CP1
PA5/U2RX_/OPO1
PA6/U2TX_/CN2
PA7/T1BK_/U2TX
PA0/T1C1N_
PA1/T1/T2/OPN0/XI/A1
PA2/T1C2N_/XO/A0
PD0/T1C3N_
PD2/T1/T2/U2TX_/CP2/A3
PD1/SWIO/T1/OPP3
PB4
PC7/T1/T2/MISO
PC6/T1/RX_/MOSI
PC5/RST/T1/T2/TX_/SCL_/SCK/CP0
PC4/T1/SDA_/MCO/A2
VDD
C14
0.1uF
P1
Header 11X2
P2
Header 10X2
XI
XO
22pF
C7
C8
Y1
24MHz
R5
R4
NC(0R)
PA1
PA2
NRST
S1
C11
0.1uF


POWER
LED
MCU
PIN
RST
R10
NC(0R)
S2
SW
F1
500mA
U1
6219P33M
Vin
Vout
GND
C12
10uF
C13
2.2uF
R1
1K
PWR
P7
R22
5.1K
R23
5.1K
+5V
VCC
R2
1K
R3
1K
LED1
LED2
P4
2HEAD
U2
CH32V006K8U
VSS
PB3/SWCK/T2/U2RX
PA5/U2RX_/OPO1
PA6/U2TX_
PA7/U2TX/RST
PA0/T1/T2
PA1/T1/T2/OPN0/XI/A1
PA2/T1/T2/U2TX_/OPP0/XO/A0
PD0/T1/TX_/SDA_/OPN1
PD2/T1/T2/U2TX_/A3
PD1/SWIO/T1/SCL_/RX_/OPP3
PB4/T1ETR_
PC7/T1C4_/T1/T2/MISO
PC6/T1C3_/T1/RX_/MOSI
PC5/T1C2_/T1/T2/TX_/SCK
PC4/T1C1_/T1/MCO/A2
VDD
C14
0.1uF
XI
XO
R5
NC(0R)
R4
NC(0R)
Y1
24MHz
C7
C8
22pF
P1
Header 11X2
P2
Header 10X2
NRST
S1
C11
0.1uF